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Precooling of electric vehicle charging stations

Abstract

Advance temperature control of charging station components is provided. A system can identify a first time at which an electric vehicle arrives at a charging station to charge the electric vehicle. The system can determine, based at least in part on the first time and a temperature of a component of the charging station, a second time prior to arrival of the electric vehicle at the charging station at which to activate a cooling system of the charging station. The system can provide an instruction to activate the cooling system at the second time to lower the temperature of the component of the charging station by the first time at which the electric vehicle arrives at the charging station to charge the electric vehicle.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10286807	12/2018	Christen	N/A	H01M 10/63
2014/0012447	12/2013	Gao	701/22	B60L 58/13
2016/0129797	12/2015	Jackson	320/109	B60L 58/26
2020/0412160	12/2019	Manzoor	N/A	B60L 50/60
2024/0326650	12/2023	Shojaei	N/A	B60L 58/12

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Background/Summary

INTRODUCTION

(1) Electric vehicles (EVs) can be powered using batteries that store energy. The EV batteries can be charged at charging stations.

SUMMARY

(2) The present disclosure is generally directed to precooling of electric charging devices, such as EV charging stations. Regulating the temperature of charging devices can allow for their operation at an increased rate of charge and reduce the amount of time to charge an electric vehicle. The temperate of components of a charging station may increase due to a variety of factors, such as the exposure to sun, ambient temperature, or prolonged or recent use. It can be beneficial to precool the charging device to allow the charging device to maintain efficient delivery of power during a charging session. The present solution provides systems and methods for precooling or otherwise controlling the temperature of EV charging station components in anticipation of arrival of EVs seeking to use the EV charging station. For example, the present solution can identify a time at which an EV is scheduled to arrive at the charging station. Based on this time and the temperature of the EV charging station components, the present solution can determine the time at which to activate the precooling of the EV charger components. In doing so, the present solution can allow for the charging station to be cooled by the time the EV arrives. By precooling the EV charger components prior to their use, this technical solution can increase the rate of charge delivered by the charging session while reducing the charge time without wasting energy, such as on idle precooling.

(3) At least one aspect is directed to a system. The system can include one or more processors,

coupled with memory. The one or more processors can identify a first time at which an electric vehicle arrives at a charging station to charge the electric vehicle. The one or more processors can determine, based at least in part on the first time and a temperature of a component of the charging station, a second time. The second time can be a time that is prior to arrival of the electric vehicle at the charging station at which to activate a cooling system of the charging station. The one or more processors can provide an instruction to activate the cooling system at the second time to lower the temperature of the component of the charging station by the first time at which the electric vehicle arrives at the charging station to charge the electric vehicle.

(4) At least one aspect is directed to a method. The method can include identifying, by one or more processors coupled with memory, a first time. The first time can be a time at which an electric vehicle arrives at a charging station to charge the electric vehicle. The method can include determining, by the one or more processors, based at least in part on the first time and a temperature of a component of the charging station, a second time. The second time can be a time prior to arrival of the electric vehicle at the charging station at which to activate a cooling system of the charging station. The method can include providing, by the one or more processors, an instruction to activate the cooling system at the second time to lower the temperature of the component of the charging station by the first time at which the electric vehicle arrives at the charging station to charge the electric vehicle.

(5) At least one aspect is directed to an electric vehicle charger. The electric vehicle charger can include one or more processors, coupled with memory. The one or more processors can receive, via a network, an indication from an electric vehicle to establish a session to charge the electric vehicle. The indication can include a state of charge of a battery of the electric vehicle and a location of the electric vehicle. The one or more processors can select, based on the state of charge of the battery and the location of the electric vehicle, a charging station from a plurality of charging stations. The one or more processors can determine, responsive to the indication and based on a distance between the electric vehicle and the charging station, a first time at which the electric vehicle arrives at the charging station to charge the electric vehicle to begin the session. The one or more processors can determine, based at least in part on the first time and a temperature of a component of the charging station, a second time prior to arrival of the electric vehicle at the charging station at which to activate a cooling system of the charging station. The one or more processors can provide an instruction to activate the cooling system at the second time to lower the temperature of the component of the charging station by the first time at which the electric vehicle arrives at the charging station to charge the electric vehicle.

(6) At least one aspect is directed to a system. The system can include one or more processors, coupled with memory. The one or more processors can access a model trained with machine learning and data comprising one or more of: values of temperature of one or more components of a charging station, values of an ambient temperature external to the charging station, metrics indicating performance of charging sessions of the charging station and a distance of an electric vehicle from the charging station. The one or more processors can identify, based on the model, a first time at which an electric vehicle arrives at a charging station to charge the electric vehicle. The one or more processors can determine, based on at least one of the first time and the temperature of the component input into the model, a second time prior to arrival of the electric vehicle at the charging station at which to activate a cooling system of the charging station. The one or more processors can provide an instruction to activate the cooling system at the second time to lower the temperature of the component of the charging station by the first time at which the electric vehicle arrives at the charging station to charge the electric vehicle.

(7) These and other aspects and implementations are discussed in detail below. The foregoing information and the following detailed description include illustrative examples of various aspects and implementations, and provide an overview or framework for understanding the nature and character of the claimed aspects and implementations. The drawings provide illustration and a

further understanding of the various aspects and implementations, and are incorporated in and constitute a part of this specification. The foregoing information and the following detailed description and drawings include illustrative examples and should not be considered as limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings are not intended to be drawn to scale. Like reference numbers and designations in the various drawings indicate like elements. For purposes of clarity, not every component may be labeled in every drawing. In the drawings:
- (2) FIG. 1 depicts an example electric vehicle connected to a charging station.
- (3) FIG. 2A depicts an example of one or more battery packs.
- (4) FIG. 2B depicts an example of one or more battery modules.
- (5) FIG. 2C depicts a cross sectional view of an example of a battery cell.
- (6) FIG. 3 is a block diagram illustrating an example architecture for a computer system that can be employed to implement elements of the systems and methods described and illustrated herein.
- (7) FIG. 4 is a block diagram illustrating an example of a system to precool an EV charging station.
- (8) FIG. 5 is a flow diagram illustrating an example method for precooling an EV charging station.
- (9) FIG. 6 is a flow diagram illustrating an example method for providing an EV charging station.

DETAILED DESCRIPTION

- (10) Following below are more detailed descriptions of various concepts related to, and implementations of, methods, apparatuses, and systems of cooling (or otherwise regulating the temperature of) of electric vehicle charging stations. The various concepts introduced above and discussed in greater detail below may be implemented in any of numerous ways.
- (11) This disclosure is generally directed to a solution for regulating temperature of an EV charging station, such as by precooling or preheating an EV charging station in order to increase the charging station's rate of charge and reduce the amount of time taken to charge an electric vehicle. As EV charging stations charge electric vehicles, the temperature of the components of the charging station used to convey power can change. For example, the temperature of the power electronics within the charging station, or the power cable, can increase. To manage the temperature of components associated with the charging session, the charging station can reduce the rate of charge (e.g., the power) delivered to the electric vehicle. However, decreasing the rate at which the EV is being charged can extend the duration of the charging session. Cooling the components of the charging station to reduce their temperature while the charge session is ongoing can use more energy than maintaining the lower temperature of the same components when such components were cool at the start of the charging session.
- (12) This technical solution can provide precooling to EV charger components prior to the arrival of the EV to be charged at the charging station. The present solution can identify the time at which the EV is to arrive at the charging station. Based on this time and the temperature of the EV charger component, the present solution can determine the time to precool the EV charger components. The present solution can issue and send an instruction to commence the cooling of the EV charger component at the time to precool the EV charger component. By precooling the EV charger component, this technical solution can increase the rate of charge delivered by the charging session, thereby reducing the duration of the charging session in an efficient manner without wasting energy on idle precooling. In the instances in which EV charger is deployed in a very cold environment, the present solution can preheat the EV charging station or its components in order to prepare for the arrival of the EV to be charged.
- (13) FIG. 1 depicts an example cross-sectional view **100** of an electric vehicle **105** installed with at least one battery pack **110**. Electric vehicles **105** can include electric trucks, electric sport utility

vehicles (SUVs), electric delivery vans, electric automobiles, electric cars, electric motorcycles, electric scooters, electric passenger vehicles, electric passenger or commercial trucks, hybrid vehicles, or other vehicles such as sea or air transport vehicles, planes, helicopters, submarines, boats, or drones, among other possibilities. Yet, it should also be noted that battery pack **110** may also be used as an energy storage system to power a building, such as a residential home or commercial building. Electric vehicles **105** can be fully electric or partially electric (e.g., plug-in hybrid) and further, electric vehicles **105** can be fully autonomous, partially autonomous, or unmanned. Electric vehicles **105** can also be human operated or non-autonomous. Electric vehicles **105** such as electric trucks or automobiles can include on-board battery packs **110**, battery modules **115**, or battery cells **120** to power the electric vehicles. The electric vehicle **105** can include a chassis **125** (e.g., a frame, internal frame, or support structure). The chassis **125** can support various components of the electric vehicle **105**. The chassis **125** can span a front portion **130** (e.g., a hood or bonnet portion), a body portion **135**, and a rear portion **140** (e.g., a trunk, payload, or boot portion) of the electric vehicle **105**. The battery pack **110** can be installed or placed within the electric vehicle **105**. For example, the battery pack **110** can be installed on the chassis **125** of the electric vehicle **105** within one or more of the front portion **130**, the body portion **135**, or the rear portion **140**. The battery pack **110** can include or connect with at least one busbar, e.g., a current collector element. For example, the first busbar **145** and the second busbar **150** can include electrically conductive material to connect or otherwise electrically couple the battery modules **115** or the battery cells **120** with other electrical components of the electric vehicle **105** to provide electrical power to various systems or components of the electric vehicle **105**.

(14) FIG. 2A depicts an example battery pack **110**. Referring to FIG. 2A, among others, the battery pack **110** can provide power to electric vehicle **105**. Battery packs **110** can include any arrangement or network of electrical, electronic, mechanical or electromechanical devices to power a vehicle of any type, such as the electric vehicle **105**. The battery pack **110** can include at least one housing **205**. The housing **205** can include at least one battery module **115** or at least one battery cell **120**, as well as other battery pack components. The housing **205** can include a shield on the bottom or underneath the battery module **115** to protect the battery module **115** from external conditions, particularly if the electric vehicle **105** is driven over rough terrains (e.g., off-road, trenches, rocks, etc.) The battery pack **110** can include at least one cooling line **210** that can distribute fluid through the battery pack **110** as part of a thermal/temperature control or heat exchange system that can also include at least one thermal component (e.g., cold plate) **215**. In some instances, the thermal component **215** may be positioned in relation to a top submodule and a bottom submodule, such as in between the top and bottom submodules, among other possibilities. The battery pack **110** can include any number of thermal components **215**. For example, there can be one or more thermal components per battery pack **110**, or per battery module **115**. The fluid (e.g., cooling) line **210** can be coupled with, part of, or independent from the thermal component **215**.

(15) FIG. 2B depicts example battery modules **115**, and FIG. 2C depicts an example cross sectional view of a battery cell **120**. The battery modules **115** can include at least one submodule. For example, the battery modules **115** can include at least one top submodule **220** or at least one bottom submodule **225**. At least one thermal component **215** can be disposed between the top submodule **220** and the bottom submodule **225**. For example, one thermal component **215** can be configured for heat exchange with one battery module **115**. The thermal component **215** can be disposed or thermally coupled between the top submodule **220** and the bottom submodule **225**. One thermal component **215** can also be thermally coupled with more than one battery module **115** (or more than two submodules **220**, **225**). The battery submodules **220**, **225** can collectively form one battery module **115**. In some examples each submodule **220**, **225** can be considered as a complete battery module **115**, rather than a submodule.

(16) The battery modules **115** can each include a plurality of battery cells **120**. The battery modules **115** can be disposed within the housing **205** of the battery pack **110**. The battery modules **115** can

include battery cells **120** that are cylindrical, prismatic, or pouch cells, for example. The battery module **115** can operate as a modular unit of battery cells **120**. For example, a battery module **115** can collect current or electrical power from the battery cells **120** that are included in the battery module **115** and can provide the current or electrical power as output from the battery pack **110**. The battery pack **110** can include any number of battery modules **115**. For example, the battery pack can have one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve or other number of battery modules **115** disposed in the housing **205**. It should also be noted that each battery module **115** may include a top submodule **220** and a bottom submodule **225**, possibly with a cold plate **215** in between the top submodule **220** and the bottom submodule **225**. The battery pack **110** can include or define a plurality of areas for positioning of the battery module **115**. The battery modules **115** can be square, rectangular, circular, triangular, symmetrical, or asymmetrical. In some examples, battery modules **115** may be different shapes, such that some battery modules **115** are rectangular but other battery modules **115** are square shaped, among other possibilities. The battery module **115** can include or define a plurality of slots, holders, or containers for a plurality of battery cells **120**.

(17) Battery cells **120** have a variety of form factors, shapes, or sizes. For example, battery cells **120** can have a cylindrical, rectangular, square, cubic, flat, pouch, or prismatic form factor. Battery cells **120** can be assembled, for example, by inserting a wound or stacked electrode roll (e.g., a jelly roll) including electrolyte material into at least one battery cell housing **230**. The electrolyte material, e.g., an ionically conductive fluid or other material, can generate or provide electric power for the battery cell **120**. A first portion of the electrolyte material can have a first polarity, and a second portion of the electrolyte material can have a second polarity. The housing **230** can be of various shapes, including cylindrical or rectangular, for example. Electrical connections can be made between the electrolyte material and components of the battery cell **120**. For example, electrical connections with at least some of the electrolyte material can be formed at two points or areas of the battery cell **120**, for example to form a first polarity terminal **235** (e.g., a positive or anode terminal) and a second polarity terminal **240** (e.g., a negative or cathode terminal). The polarity terminals can be made from electrically conductive materials to carry electrical current from the battery cell **120** to an electrical load, such as a component or system of the electric vehicle **105**.

(18) The battery cell **120** can include a lithium-ion battery cell, for example. In lithium-ion battery cells, lithium ions can transfer between a positive electrode and a negative electrode during charging and discharging of the battery cell. For example, the battery cell anode can include lithium or graphite, and the battery cell cathode can include a lithium-based oxide material. The electrolyte material can be disposed in the battery cell **120** to separate the anode and cathode from each other and to facilitate transfer of lithium ions between the anode and cathode. It should be noted that battery cell **120** can also take the form of a solid state battery cell developed using solid electrodes and solid electrolytes. Yet further, some battery cells **120** can be solid state battery cells and other battery cells **120** can include liquid electrolytes for lithium-ion battery cells.

(19) The battery cell **120** can be included in battery modules **115** or battery packs **110** to power components of the electric vehicle **105**. The battery cell housing **230** can be disposed in the battery module **115**, the battery pack **110**, or a battery array installed in the electric vehicle **105**. The housing **230** can be of any shape, such as cylindrical with a circular (e.g., as depicted), elliptical, or ovular base, among others. The shape of the housing **230** can also be prismatic with a polygonal base, such as a triangle, a square, a rectangle, a pentagon, and a hexagon, among others.

(20) The housing **230** of the battery cell **120** can include one or more materials with various electrical conductivity or thermal conductivity, or a combination thereof. The electrically conductive and thermally conductive material for the housing **230** of the battery cell **120** can include a metallic material, such as aluminum, an aluminum alloy with copper, silicon, tin, magnesium, manganese, or zinc (e.g., aluminum 1000, 4000, or 5000 series), iron, an iron-carbon

alloy (e.g., steel), silver, nickel, copper, and a copper alloy, among others. The electrically insulative and thermally conductive material for the housing **230** of the battery cell **120** can include a ceramic material (e.g., silicon nitride, silicon carbide, titanium carbide, zirconium dioxide, beryllium oxide, and among others) and a thermoplastic material (e.g., polyethylene, polypropylene, polystyrene, polyvinyl chloride, or nylon), among others.

(21) The battery cell **120** can include at least one anode layer **245**, which can be disposed within the cavity **250** defined by the housing **230**. The anode layer **245** can receive electrical current into the battery cell **120** and output electrons during the operation of the battery cell **120** (e.g., charging or discharging of the battery cell **120**). The anode layer **245** can include an active substance. The active substance can include, for example, an activated carbon or a material infused with conductive materials (e.g., artificial or natural Graphite, or blended), lithium titanate (Li.sub.4Ti.sub.5O.sub.12), or a silicon-based material (e.g., silicon metal, oxide, carbide, pre-lithiated).

(22) The battery cell **120** can include at least one cathode layer **255** (e.g., a composite cathode layer compound cathode layer, a compound cathode, a composite cathode, or a cathode). The cathode layer **255** can be disposed within the cavity **250**. The cathode layer **255** can output electrical current out from the battery cell **120** and can receive electrons during the discharging of the battery cell **120**. The cathode layer **255** can also release lithium ions during the discharging of the battery cell **120**. Conversely, the cathode layer **255** can receive electrical current into the battery cell **120** and can output electrons during the charging of the battery cell **120**. The cathode layer **255** can receive lithium ions during the charging of the battery cell **120**.

(23) The battery cell **120** can include an electrolyte layer **260** disposed within the cavity **250**. The electrolyte layer **260** can be arranged between the anode layer **245** and the cathode layer **255** to separate the anode layer **245** and the cathode layer **255**. The electrolyte layer **260** can transfer ions between the anode layer **245** and the cathode layer **255**. The electrolyte layer **260** can transfer cations from the anode layer **245** to the cathode layer **255** during the operation of the battery cell **120**. The electrolyte layer **260** can transfer anions (e.g., lithium ions) from the cathode layer **255** to the anode layer **245** during the operation of the battery cell **120**.

(24) The electrolyte layer **260** can include or be made of a liquid electrolyte material. The liquid electrolyte material can include a lithium salt dissolved in a solvent. The lithium salt for the liquid electrolyte material for the electrolyte layer **260** can include, for example, lithium tetrafluoroborate (LiBF.sub.4), lithium hexafluorophosphate (LiPF.sub.6), and lithium perchlorate (LiClO.sub.4), among others. The solvent can include, for example, dimethyl carbonate (DMC), ethylene carbonate (EC), and diethyl carbonate (DEC), among others. The electrolyte layer **260** can include or be made of a solid electrolyte material, such as a ceramic electrolyte material, polymer electrolyte material, or a glassy electrolyte material, or among others, or any combination thereof. The ceramic electrolyte material for the electrolyte layer **260** can include, for example, lithium phosphorous oxy-nitride (Li.sub.xPO.sub.yN.sub.z), lithium germanium phosphate sulfur (Li.sub.10GeP.sub.2S.sub.12), Ytria-stabilized Zirconia (YSZ), NASICON (Na.sub.3Zr.sub.2Si.sub.2PO.sub.12), beta-alumina solid electrolyte (BASE), perovskite ceramics (e.g., strontium titanate (SrTiO.sub.3)), among others. The polymer electrolyte material (e.g., a hybrid or pseudo-solid state electrolyte) for electrolyte layer **260** can include, for example, polyacrylonitrile (PAN), polyethylene oxide (PEO), polymethyl-methacrylate (PMMA), and polyvinylidene fluoride (PVDF), among others. The glassy electrolyte material for the electrolyte layer **260** can include, for example, lithium sulfide-phosphor pentasulfide (Li.sub.2S—P.sub.2S.sub.5), lithium sulfide-boron sulfide (Li.sub.2S—B.sub.2S.sub.3), and Tin sulfide-phosphor pentasulfide (SnS—P.sub.2S.sub.5), among others.

(25) FIG. 3 depicts an example block diagram of an example computer system **300**. The computer system or computing device **300** can include or be used to implement a data processing system or its components. The computing system **300** includes at least one bus **305** or other communication

component for communicating information and at least one processor **310** or processing circuit coupled to the bus **305** for processing information. The computing system **300** can also include one or more processors **310** or processing circuits coupled to the bus for processing information. The computing system **300** also includes at least one main memory **315**, such as a random access memory (RAM) or other dynamic storage device, coupled to the bus **305** for storing information, and instructions to be executed by the processor **310**. The main memory **315** can be used for storing information during execution of instructions by the processor **310**. The computing system **300** may further include at least one read only memory (ROM) **320** or other static storage device coupled to the bus **305** for storing static information and instructions for the processor **310**. A storage device **325**, such as a solid state device, magnetic disk or optical disk, can be coupled to the bus **305** to persistently store information and instructions.

(26) The computing system **300** may be coupled via the bus **305** to a display **335**, such as a liquid crystal display, or active matrix display, for displaying information to a user such as a driver of the electric vehicle **105** or other end user. An input device **330**, such as a keyboard or voice interface may be coupled to the bus **305** for communicating information and commands to the processor **310**. The input device **330** can include a touch screen display **335**. The input device **330** can also include a cursor control, such as a mouse, a trackball, or cursor direction keys, for communicating direction information and command selections to the processor **310** and for controlling cursor movement on the display **335**.

(27) The processes, systems and methods described herein can be implemented by the computing system **300** in response to the processor **310** executing an arrangement of instructions contained in main memory **315**. Such instructions can be read into main memory **315** from another computer-readable medium, such as the storage device **325**. Execution of the arrangement of instructions contained in main memory **315** causes the computing system **300** to perform the illustrative processes described herein. One or more processors in a multi-processing arrangement may also be employed to execute the instructions contained in main memory **315**. Hard-wired circuitry can be used in place of or in combination with software instructions together with the systems and methods described herein. Systems and methods described herein are not limited to any specific combination of hardware circuitry and software.

(28) Although an example computing system has been described in FIG. 3, the subject matter including the operations described in this specification can be implemented in other types of digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them.

(29) FIG. 4 illustrates an example system **400** of the present solution for temperature control or regulation (e.g., precooling) of an EV charging station in anticipation of an EV set to arrive to the charging station. As shown in FIG. 4, one or more charging stations (CS) **405** can be deployed at one or more locations. Charging stations **405** can communicate with one or more EVs **105** via one or more networks **101**. Each charging station **405** can include one or more computer systems **300**, one or more components **410**, such as charging cables **160** or power electronics **165**, which can be used to charge an EV **105**. Charging station **405** can include one or more charge controllers **450** that can include one or more session managers **455** to manage one or more sessions **475**. Components **410** of the CS **405** can be used for charging EVs **105** and can be cooled by one or more coolers **420**. Cooler **420** can include a heater that can be used for regulating temperature. For example, cooler **420** can include an electrical heating component or a heating device that can raise the temperature of one or more components **410** in order to place the component **410** at desired or target temperature range. The desired or target temperature range can be a temperature range that allows performance of the EV charging station **405** at a particular power level. For example, if an EV charging station **405** is deployed in very cold environment, the temperate can be raised in order to reach a temperature range that allows for charging of EV **105** at a highest power level. In such

instances, DPS **480** can activate a heater or a heating unit in a cooler **420** to increase the temperature of the component **410** and place the EV charging station **405** at a desired target level. (30) Each cooler **420** can include one or more cooling controller **445** and one or more temperature control (e.g., cooling) functions **425**. Each cooling function **425** can include one or more inputs **430**, one or more outputs **435** and one or more instructions **440**. Cooling controller **445** can control temperature regulation of EV charging station **405** or its components **410**, including precooling or preheating components **410**. CS **405** can include one or more network interfaces **460** and one or more sensors **465**. CS **405** can communicate with a data processing system **480**, via network **101**. Data processing system (DPS) **480** can include one or more computer systems **300**, cooling functions **425**, charge controllers **450**, network interfaces **460** and indications **415**. Across the network **101**, each EV **105** can include one or more network interfaces **460** for communicating with the charging stations **405**. System **400** can utilize one or more CSs **405** or one or more servers **480** to precool or preheat components **410** used for charging EVs **105** prior to the arrival of the EV **105** to the CS **405** location or site, based on a time when the EV **105** arrives at the CS **405** and the temperature of component **410**.

(31) A charging station **405** can include any combination of hardware and software for providing electricity or otherwise electrically charging one or more batteries of one or more EVs **105**. CS **405** can be a bidirectional charging station and can include any combination of hardware and software for providing and drawing power or energy or otherwise charging and discharging one or more batteries of the EV **105**, including battery packs **110**, battery modules **115** or battery cells **120**. CS **405** can include scripts, functions and computer code stored in memory and executed or operating on one or more processors to implement any functionality of the CS **405**. CS **405** can include a computer system **300**, including one or more processors **310** and memories **315**, **320** and **325**, each of which can store computer code, scripts, functions and instructions to implement any functionality of CS **405**. CS **405** can include electrical and power circuitry, control logic or circuits, power electronics, power supply circuitry, energy storage devices, such as batteries, and other hardware for storing, controlling, modulating or otherwise managing power, energy or electricity provided to, or drawn from, EVs **105**. CS **405** can include electric vehicle charging equipment that can include a power and control box and power cord or a cable. CS **405** can include circuitry for converting alternating current (AC) to direct current (DC), such as an AC-DC converter. CS **405** can include DC-AC converters or DC-DC converters.

(32) Charging station **405** can include one or more bidirectional chargers which can be electrically coupled to an electrical grid. The CS **405** can draw electricity from the grid to charge EVs **105** and can receive electricity from the batteries (e.g., **110**, **115** or **120**) of the EV **105** and provide this electricity to the electrical grid. Charging station **405** can include or configure any level of EV charging equipment, such as level-1, level-2, level-3 or any other charging equipment level known or used in the industry. Charging station **405** can be set to operate, including provide or draw electricity, at any voltage, current or power level. For example, CS **405** can provide electricity to EVs or draw power from EVs **105** at any voltage level, such as 120V, 208-240V or 400-900V. Similarly, CS **405** can provide electricity to EVs **105** or draw electricity from EVs **105** at any power levels, such as anywhere between 5 and 800 kW, such as 5 kW, 10 kW, 20 kW, 30 kW, 50 kW, 80 kW, 100 kW, 150 kW, 220 kW, 250 kW, 300 kW, 350 kW, 500 kW, or more. Higher power throughput by CS **405** can result in increased thermal energy dissipation and increase the temperature of some of the components of the CS **405**. As the temperature of a component increases, the CS **405** can decrease the power output or charge rate to or from the EV **105** to maintain the temperature of the component at or below an established thermal threshold or recommended level for the component **410**. Charging station **405** can utilize coolers **420** to cool down components of the CS **405** and maintain the power, voltage or current throughput at a higher level while satisfying the thermal threshold for the component **410**.

(33) Components **410** of the CS **405** can include any component, part, subsystem or system of the

CS **405** generating thermal energy. Components **410** can include any one or more components of a charging station **405**, such as a control box housing power circuitry or a power cord or a cable for charging the EV **105**. Components **410** can include power electronics **165** which can include any circuitry, including for example, AC-DC converters, DC-DC converters, DC-AC converters, any combination of power transistors, capacitors, inductors, resistors, diodes, switches, transformers, relays and other electrical or electronic components to form structures, such as half and full bridge circuits, rectifiers, filters, multi-function circuits, single or multi-stage chargers with resonant half-bridge converts utilizing one or more inductors and one or more capacitors, such as the LLC converters and single or multi-directional DC-DC converters. Components **410** can include memory, such as **315**, **320** or **325** memories, that can store scripts, computer code or instructions to be accessed or executed by electronic microcontrollers or devices, such as processors **310**.

Components **410** can include one or more energy storage systems, including batteries. Components **410** can include lithium-ion, solid-state or any other type and form of battery type.

(34) Component **410** can include a power cable **160**, which can also be referred to as the power cord **160**. Power cable or cord **160** can be attached to or coupled with power electronics **165** and cooled by a cooler **420** device. The power cable **160** can include one or more electrical conductor wires or lines, including lines or wires for high power throughput as well as electronic or electrical signals. Component **410** can include a power plug for plugging into an EV **105**. Component **410** can include wires or lines for conducting high power, high voltage or high current between EV **105** and CS **405**. Component **410** can include one or more wires or lines for conducting analog or digital communication signals between the EV **105** and CS **405**. Component **410** can include communication circuitry, such as processors or microcontrollers for receiving, sending and processing communication via the wires or lines. Component **410** can include a threshold, such as a threshold temperature for which the component **410** is rated.

(35) Indications **415** can include any type and form of data, information or notification that can be used for cooling or precooling of CS **405** or its components **410**. Indication **415** can be information or data used for making determinations or decisions with respect to the precooling of charging station **405** components **410**. Indication **415** can be any information received via a network interface **460**, any information determined by CS **405** or its components **410** or any information measured by or received from sensors **465**. Indication **415** can include any data or information about CS **405**, environment of the CS **405**, or the environment of EV **105**.

(36) Indication **415** can include, for example, an amount of power to be provided to the EV **105**, an amount of power to be drawn from the EV **105**, information about a temperature of a component **410** or a EV **105**, including, e.g., temperature of an EV battery **110**. Indication **415** can include information or indication of a temperature of a component **410**, which can be received over a network **101** via network interface **460** or be measured from a sensor **465** and provided to the CS **405**. For example, indication **415** can include temperature measurements of a power plug, power cord or cable, power electronics, specific electrical circuits of one or more power supplies, converters or transformers, or a temperature of control box of CS **405**. Indication **415** can include a temperature of the ambient or surrounding environment in which the CS **405** is located or deployed. Indication **415** can include a temperature threshold for a particular component **410**, such as a maximum rated temperature for a component **410**.

(37) Indication **415** can include information about an EV **105**. For example, indication **415** can include information about the location of the EV **105**, such as global positioning system (GPS) location, an address at which the EV **105** is located or a distance of the EV **105** from the CS **405**. Indication **415** can include speed or velocity of the EV **105**. Indication **415** can include estimate of average speed EV **105** will travel to CS **405**. Indication **415** can include amount of power that EV **105** needs to receive from CS **405** or that an EV **105** needs to provide to CS **405**. Indication **415** can include a state of charge of a battery pack **110** of EV **105**.

(38) Indication **415** can include a request received over a network **101** to book or reserve a CS **405**

for an arriving EV **105**. Indication **415** can include an indication or a request to establish a connection or a session **475** with CS **405**. Indication **415** can identify a driver of an EV **105** or the EV **105** itself. Indication **415** can include information about an account of the driver of EV **105**, such as a payment account for the driver to pay for the EV **105** charging station services. Indication **415** can include information about the temperature of the battery of the EV **105**, rate of charge EV **105** requests and/or information about duration of time during which EV **105** is going to be connected to the CS **405** to complete its charging.

(39) Coolers **420** can include any thermal management system for cooling CS **405** or its components **410**. Cooler **420** can include any one or more devices or systems for dissipating thermal energy, such as electrical cooling systems, air cooling systems or water cooling systems. Cooler **420** can include a water cooling loop or loops that can flow through a path to take the heat from components **410** and deliver the heat to the heat exchanger unit that takes the heat away from the water and sends the cooled water back into the loop to continue looping and removing thermal energy. Cooler **420** can include any combination of lines or pipes for conducting or flowing liquid coolant, one or more fluid pumps, heat exchangers, cold plates, air or water fans, heat sinks, active and passive cooling systems and structures, heat pipes, Peltier systems, radiators, natural or forced convection cooling devices, refrigerators or any other components that can be used for cooling any one or more components **410**.

(40) Cooler **420** can be thermally interfaced with any components **410** of CS **405** to be cooled. Cooler **420** can be in thermal contact or coupling with any one or more of components **410**, such as power cords or cables, power plugs, power electronics or circuits or control box of the CS **405**. Cooler **420** can be in a thermal contact with any electrical, mechanical, electromechanical, electrochemical or other components or systems of a CS **405**, including AC-DC converters, DC-DC converters, energy storage systems, transformers, processors, transistors, relays or any other components **410** used in a CS **405**. Cooler **420** can be coupled to or operate on any one or more rates of cooling as determined by cooling functions **425**.

(41) Inputs **430** can include any information input into one or more cooling functions **425**. Input **430** can include a reading or a determination, including a reading or a measurement from a sensor **465**. Input **430** can include sensor **465** reading corresponding to temperature, pressure, vibration, light, or any other sensor **465** reading. Input **430** can include a temperature reading of a component **410**, CS **405** ambient temperature, EV **105** or battery pack **110** of the EV **105**. Input **430** can include time, such as time at which EV **105** is to arrive at the CS **405** or a location at which one or more CSs **405** are deployed. Input **430** can include any indication **415**. Input **430** can include a distance between EV **105** and CS **405**. Input **430** can include any data of a CS **405**, such as the status of a CS **405**, amount of energy stored at a battery of CS **405**, health status of CS **405**, metrics of sessions **475** or metrics performance data, rates of charge or discharge available to the CS **405**, a rate of cooling at the CS **405**, a notification concerning a reservation for an EV **105** to be charged at the CS **405**. Input **430** can include an amount of power to charge to, or discharge from, the battery pack **110** of EV **105** or information about EV **105** or the driver of **105**. Input **430** can include metrics of a session **475**, such as metrics of the session **475** performance.

(42) Outputs **435** can include any information determined by the cooling function **425** based one or more inputs **430** input into the cooling function **425**. Output **435** can include the time at which to turn on one or more coolers **420**. The one or more coolers can be for one or more components **410**, such as power electronics control box or power cord or a cable. Output **435** can include the time at which the EV **105** is to arrive at the CS **405**. Output **435** can include an amount of time that EV **105** is going to take to arrive at CS **405**. Output **435** can include information to start a session **475** for an EV **105**. Output **435** can include an amount of power to charge a battery pack **110** of EV **105**. Output **435** can include an amount of power to draw from the battery pack **110** of EV **105**. Output **435** can include a temperature, such as a temperature of one or more components **410** at which the rate of charge from CS **405** can increase to a predetermined rate of charge.

(43) Cooling functions **425** can include any logic or functionality for management of precooling of components **410** of CS **405**. Cooling function **425** can include any combination of hardware and software, including any scripts, functions and computer code stored in memory and executed or operating on one or more processors to implement any functionality of the cooling functions **425**. Cooling function **425** can include functionality for determining timing and amount of cooling to be provided to different components **410**. Cooling function **425** can include functionality for determining a rate of charge to provide to EV **105** or draw from EV **105**, a rate of cooling to provide to a component **410**, a temperature to which to cool a component **410**, a temperature of a component **410** at which a particular rate of charge or discharge can be provided, a time when an EV **105** arrives to the CS **405**, a duration of time that it takes for EV **105** to arrive at CS **405** or an amount of power to provide to EV **105** or draw from EV **105**.

(44) Cooling function **425** can include the functionality to determine the time of arrival of the EV **105**. Cooling function **425** can utilize the EV's location from an indication **415** to determine the time at which the EV **105** is to arrive at the given CS **405**. Cooling function **425** can utilize information about the speed of the EV **105** to determine the time of EV's arrival. Cooling function **425** can use traffic information or data to determine the time of EV's arrival.

(45) Cooling function **425** can include functionality for determining any output **435** based on input **430**. For example, cooling functionality **425** can determine the time at which to turn on, or activate, precooling of a component **410**. Cooling function **425** can determine the rate of cooling of different coolers **420** are to be operated or which coolers **420** to activate and which not to activate. Cooling function **425** can determine to activate one or more coolers **420** at one or more times, at one or more cooling rates, while activating other coolers **420** at other rates of cooling. Cooling function **425** can determine temperature to which to reduce the temperature of one or more components **410** to cool by the cooler **420**. Cooling function **425** can determine or predict the time at which the EV **105** is to arrive at CS **405** or time that it takes for the EV **105** to arrive. Cooling function **425** can determine the amount of cooling to provide based on the rate of charge. Cooling function can determine or establish a session for the EV **105**.

(46) Cooling function **425** can make determinations using a model, which can include a machine learning (ML) model or an artificial intelligence (AI) model. The model can make any determinations of outputs **435**, based on inputs **430** that can be input into the model. The model can make determinations using, or based on, any type of ML technique, including, for example, supervised learning, unsupervised learning or reinforcement learning. The model can use or include functions such as linear regression, logistic regression, a decision tree, support vector machine, Naïve Bayes, k-nearest neighbor, k-means, random forest, dimensionality reduction function, or gradient boosting functions. Cooling function **425** can include any functionality for training an ML or AI model of cooling function **425**.

(47) Cooling functions **425** can issue instructions **440**. For example, the cooling controller **445** can execute a cooling function **425** to generate and issue an instruction **440**. Instructions **440** can include any instructions based on output **435**. Instructions **440** can include any instructions implementing output of **435**. For example, instruction **440** can be an instruction to turn on a cooler **420**, such as a cooling loop for a particular component **410**. For example, instruction **440** can include an instruction to activate a cooling loop for a power cable or a power cord that plugs into EV **105**. Instruction **440** can include an instruction to activate a cooling loop for a power electronics box or module that includes circuitry managing power delivered to EV **105** or drawn from EV **105**. Instruction **440** can include an instruction to set a rate of cooling, a rate of charge, a temperature to which to cool a component **410**, a time at which to activate a cooler **420**, which cooler **420** to activate or to establish a session for EV **105**.

(48) Cooling controllers **445** can include any combination of hardware and software for controlling coolers **420**, including valves, relays and control circuits activating, deactivating and controlling rate of cooling by coolers **420**. Cooling controller **445** can include scripts, functions and computer

code stored in memory and executed or operating on one or more processors to implement any functionality of the cooling controller **445**. Cooling controller **445** can include control circuitry or devices for activating and deactivating coolers **420**, including cooling loops for particular components **410**, setting or controlling a rate of cooling by a cooler **420** or maintaining cooling of a component **410** until temperature of that component **410** meets or exceeds a threshold temperature. (49) Charge controllers **450** can include any combination of hardware and software for controlling or managing charge or amount of power provided to EV **105**. Charge controller **450** can include scripts, functions and computer code stored in memory and executed or operating on one or more processors to implement any functionality of the charge controller **450**. Charge controller **450** can set and control a rate of charge to provide to EV **105** or a rate of charge to draw from an EV **105**. Charge controller **450** can set a power level, a voltage level or a current level for an EV charge, in either direction. Charge controller **450** can set up and maintain sessions **475** using session managers **455**. Charge controller **450** can maintain a rate of charge via a power cable between the CS **405** and EV **105** based on temperature of CS **405**, such as temperature of component **410**, such as a power or charging cable or power electronics box or circuit. Charge controller **450** can set, establish or maintain a rate of charge based on a temperature threshold for a component **410**, such as for example maximum rated temperature for the component **410**. Charge controller **450** can set, establish or maintain a rate of charge based on a temperature or a temperature range at which the component **410** can output at a higher power range. For example, charge controller **450** can increase the rate of charge at a higher power level, based on the temperature of a component **410** (e.g., power cable or power electronics box) reaching or exceeding below a threshold temperature for that component **410** and at that rate of charge.

(50) Session managers **455** can include any combination of hardware and software for managing sessions **475**. Session manager **455** can include scripts, functions and computer code stored in memory and executed or operating on one or more processors to implement any functionality of the session manager **455**. Session manager **455** can establish a connection and/or session with an EV **105**. Session manager **455** can communicate with EV **105** to set, establish, monitor or maintain the session **475**, including the power received from or delivered to EV **105**, rate of charge to and from EV **105**, temperature of the EV battery **110**, state of charge or state of health of EV battery **110**, temperature of components **410** corresponding to the session **475** for the particular EV **105** as well as any settings or configurations for the particular session **475**, such as the timing for activating cooler **420** for the CS **405** that corresponds to the session **475** and/or EV **105**, the rate of cooling or rate of charge.

(51) Sessions **475** can include any sessions of the CS **405** for providing charge to or from EV **105**. Session **475** can include a session with any number of EVs **105**. Session **475** can include a session with a particular EV **105**, such as an EV **105** from which CS **405** has received an indication **415**. Session **475** can be established to set up any configurations for the interaction of the CS **405** with EV **105**, such as establishing, setting and/or implementing the timing of precooling (e.g., activating one or more coolers **420**), amount of cooling to provide, the component **410** to cool, the particular cooler **420** to activate, the rate of cooling to provide, the amount of charge to provide or receive, the rate of charge or a temperature to achieve per cooling and/or charging operation. Session **475** can include any inputs **430**, outputs **435** and instructions **440** for a particular EV **105**. Session **475** can include or process EV **105** driver's account or information, such as a payment processing account or information. Session **475** can include metrics

(52) Network interfaces **460** can include any combination of hardware and software for communicating via a network **101**. Network interface **460** can include scripts, functions and computer code stored in memory and executed or operating on one or more processors to implement any functionality of the network interface **460**. Network interface **460** can include functionality for communicating via network **101** using any network communication protocol such as Transmission Control Protocol (TCP)/Internet Protocol (IP), user datagram protocol (UDP), or

any other communication protocol used for communicating over a network **101**. Network interface **460** can include network communication ports and hardware for receiving and sending communications over the network **101**.

(53) Sensors **465** can include any device that detects or measures physical properties of either a CS **405** or an EV **105**. Sensor **465** can include a temperature sensor, such as a thermocouple, resistance temperature detector (RTD), thermistor, or semiconductor based integrated circuit for measuring temperature. Sensor **465** can include a voltage, power or resistance sensor, device or a circuit. Sensor **465** can include a pressure sensor, vibration sensor, proximity sensor, velocity sensor, accelerometer, light sensor, smoke sensor, gas sensor, infra-red sensor, water flow sensor, humidity sensor, or any other type of sensor or detector.

(54) Data processing system (DPS) **480** can include any combination of hardware and software for processing information or data concerning cooling of EVs **105** and communicating information with the CS **405** and EVs **105** over the network **101**. DPS **480** can comprise the functionality for managing cooling operations of one or more remote CSs **405**, including performing any functionality of cooling functions **425** and charge controllers **450** on behalf of any remote charging stations **405**. Data processing system **480** can operate on a remote server, a server farm, a virtual machine or a cloud computing system comprising one or more servers or located at a data center. DPS **480** can include or be coupled to a computer system **300**, which can include one or more processors **310** that can be coupled with memory (e.g., **315**, **320**, **325**), each of which can store computer code, scripts, functions and instructions to implement any functionality of the DPS **480** or the CS **405**. DPS **480** can include scripts, functions and computer code stored in memory and executed or operating on one or more processors perform processing related to precooling of an EV **105** on a CS **405**. For example, DPS **480** can have access to or include one or more cooling functions **425**, charge controllers **450** and indications **415**. DPS **480** can perform the functionalities of the cooling functions **425** and charge controllers **450** remotely and receive inputs **430** from the CS **405** and communicate instructions **440** to the CS **405**. DPS **480** can perform all determinations corresponding to precooling and cooling control that can be performed by CS **405**.

(55) Data collector **485** can include any combination of hardware and software for collecting and storing all data from various charging stations **405** and various EVs **105**. Data collector **485** can be designed, constructed and operational to receive data, such as in a real-time data stream, a data feed, a batch upload, or responsive to a request, query or poll for data. Data collector **485** can receive and store data or information, such as indications **415** from EVs **105** or CSs **405**. Data collector **485** can receive and store inputs **430**, outputs **435** and instructions **440**. Data collector **485** can include a database with data structures for one or more or each CS **405** and one or more or each EV **105**. Data collector **485** can provide data to the DPS **480** for processing, to allow for the DPS **480** to operate on the inputs **430** and indications **415** and provide outputs **435** and instructions **440** to the remote CSs **405**.

(56) Charging station selector **490** can include any combination of hardware and software for selecting CSs **405** for an EV **105**. Charging station selector **490** can include the functionality of identifying which CSs **405** are most suitable to service an EV **105**. For example, charging station selector **490** can include the functionality to determine locations of closest CSs **405** for an EV **105**, based on an indication of EV **105**, such as EV's location. Charging station selector **490** can select for the EV **105** the closest EV **105**. Charging station selector **490** can include the functionality to determine which CS **405** is available and then select for an EV **105** the closest available CS **405**. Charging station **405** can include the functionality to determine which CS **405** has components **410** whose temperature is below a threshold (e.g., sufficiently cooled) and can select for the EV **105** the closest CS **405** whose components **410** are sufficiently precooled. For example, if a first charging station **405** and a second charging station **405** are both located the same distance from an EV **105** and both the first and second charging stations are available to charge the EV **105**, the charging station selector **490** can select the charging station having components **410** that are cooler (e.g.,

have lower temperature). For example, the first charging station **405** may be 5% cooler than the second charging station **405**, in which case the charging station selector **490** can select the first charging station **405**. The data processing system **480** can provide an instructions or other indication to the EV **105** to direct and cause the EV **105** to arrive at the first charging station **405**. In some cases, the charging station selector **490** can instruct the first charging station **405** to display a visual indication (e.g., a light, color, text, or image), or an auditory indication (e.g., an alarm, beep, or other sound) to cause the EV **105** to arrive at the selected CS **405**. The charging station selector **490** can be configured with one or more selection techniques, policies, heuristics, or rules to select charging stations **405** for EVs **105**.

(57) The present solution can include a system that can be coupled to or include one or more processors **310** that can be coupled with memory (e.g., **315**, **320**, **325**). The one or more processors can be operating on a data processing system **480** or on a charging station **405**. The one or more processors **310** can identify a first time (e.g., **430**) at which an electric vehicle **105** arrives at a charging station **405** to charge the electric vehicle **105**. The one or more processors **310** can determine, based on any one or more inputs **430** input into cooling functions **425**, one or more outputs **435**. For example, the one or more processors can determine, based at least in part on the first time (e.g., **430**) and a temperature of a component (e.g., **430**) of the charging station **405**, a second time (e.g., **435**) prior to arrival of the electric vehicle **105** at the charging station **405**, at which to activate a cooling system (e.g., **420**) of the charging station (e.g., **405**). The one or more processors **310** can issue or provide one or more instructions **440** to activate a cooling system **420**. For example, the one or more processors (e.g., **310**) can provide an instruction (**440**) to activate the cooling system (**420**) at the second time (**435**) to lower the temperature of the component (**410**) of the charging station by the first time at which the electric vehicle **105** arrives at the charging station **405** to charge the electric vehicle **105**.

(58) The one or more processors **310** can receive, via a network **101**, an indication **415** from the electric vehicle **105** to establish a session **475** to charge the electric vehicle **105**. The one or more processors **310** can determine, responsive to the indication **415** and based on a distance between the electric vehicle and the charging station (e.g., **430**), the first time (e.g., **435**) at which the electric vehicle arrives at the charging station to begin the session. The one or more processors **310** can receive, via a network **101**, an indication **415** of a location (e.g., **430**) of the electric vehicle **105**. The one or more processors **310** can select, based on the location (e.g., **430**), the charging station **405** from a plurality of charging stations **405**. For example, a cooling function **425** can utilize an artificial intelligence model to determine an estimated time of arrival for the EV **105**. This determination can be made, by a DPS **480**, based on one or more indications **415**, which can indicate EV's location, the location of the CS **405**, the speed at which the EV **105** is traveling, the current traffic conditions, the weather conditions and any other data that can be used to determine the most likely time it will take for the EV **105** to arrive at a given CS **405**. The one or more processors **310** can establish, based on a distance (e.g., **430**) between the selected charging station **405** and the electric vehicle **105**, a session **475** to begin to charge the electric vehicle **105** at the first time (e.g., **430**).

(59) The one or more processors **310** can receive, via a sensor **465**, an indication **415** of a first value of the temperature (e.g., **430**) of the component **410**. The component **410** can include at least one of a power cable (e.g., power cable **160** depicted in FIG. 1) or power electronics (e.g., power electronics **165** depicted in FIG. 1). The one or more processors **310** can compare the first value (e.g., **430**) with a threshold established for the component **410**. For example, the first value can correspond to the temperature of the power cable **160** or power electronics **165** in the CS **405**. The power cable **160** or power electronics **165** can each have their own rated threshold temperature. The one or more processors **310** can compare the first value of the temperature against the rate threshold temperature for the component **410**. Each of the components **410**, such as the power cable **160** or power electronics **165**, can have different threshold temperatures for operating at

different power levels. For example, there can be a temperature threshold for operating a component **410** at a higher power level, which can be different from a threshold temperature when the component is operated at a lower power level.

(60) The one or more processors **310** can determine, based at least in part on a comparison of the first value (e.g., **430**) with the threshold, such as the rated temperature threshold or the power level threshold, to activate the cooling system (e.g., **420**) at the second time (e.g., **435**). The one or more processors **315** can determine the second time in response to determining that cooling the component to a particular threshold is going to take additional time. The one or more processors **310** can receive, via a sensor **465**, an indication **415** of an ambient temperature (e.g., **430**) that is external to the charging station **405**. The one or more processors **310** can determine, based at least in part on the first time (e.g., **430**), the temperature (e.g., **430**) of the component **410**, and the ambient temperature (e.g., **430**) external to the charging station **405**, to activate the cooling system **420** at the second time (e.g., **435**).

(61) The one or more processors **310** can receive, from the electric vehicle **105** via a network **101**, an indication (e.g., **415**, **430**) of a state of charge of a battery (e.g., **110**, **115**, **120**) of the electric vehicle **105** and a location of the electric vehicle (e.g., **430**). The one or more processors **310** can select, based on the state of charge of the battery and the location of the electric vehicle, the charging station **405** from a plurality of charging stations **405**. For example, a data processing system **480** can utilize the charging station selector **490** to select the charging station **405**, based on the input from the EV **105** identifying the state of battery charge and location of the EV. The one or more processors **310** can predict, based on a distance (e.g., **430**, **435**) between the selected charging station **405** and the electric vehicle **105**, the first time (e.g., **435**) at which the electric vehicle **105** arrives at the charging station **405**. For example, the data processing system **480** can identify from its data collector **485** charging stations **405** in the area in which EV **105** is seeking to charge its battery. Upon selecting the charging station **405** that is available to the EV **105**, the data processing system **480** can use the distance between the EV **105** and the selected CS **405** to determine the time of EV's arrival to the CS **405**. DPS **480** can make the determination of the time of arrival based on the traffic data, weather data as well as road conditions the EV is expected to travel to the CS **405**.

(62) The one or more processors **310** can receive, from the electric vehicle **105** via a network **101**, an indication **415** of a state of charge of a battery (e.g., **110**, **115**) of the electric vehicle **105**. The one or more processors **310** can determine, based on the state of charge, an amount of power to charge (e.g., **435**) the battery (e.g., **110**, **115**, **120**) of the electric vehicle **105** and establish, based on the amount of power to charge the battery, to activate the cooling system **420** at the second time (e.g., **435**). The one or more processors **310** can receive a first value (e.g., **415**, **430**) of the temperature of the component **410** measured via a sensor **465**. The one or more processors **310** can determine, based on the amount of power to charge the battery (e.g., **110**, **115**, **120**), a second value (e.g., **435**) of the temperature for the component **410** less than the first value (e.g., **435**) that increases a rate of charge of the charging station **405** relative to the first value of the temperature. The one or more components **310** can determine the second time (e.g., **435**) at which to activate the cooling system **420** based on the first value (e.g., **435**) of the temperature, the second value (e.g., **435**) of the temperature, and the first time of arrival (e.g., **430**). Determination can be made by charge controller **450** using a session **475** of the EV **105** associated with the first time of arrival (e.g., **430**).

(63) The one or more processors **310** can establish a rate of cooling of the cooling system **420** based on the first value (e.g., **435**) of the temperature, the second value (e.g., **435**) of the temperature, and the first time of arrival (e.g., **430**) and provide the instruction **440** to activate the cooling system **420** comprising an instruction to cool at the established rate of cooling. The one or more processors **310** can instruct the charging station **405** to present a notification at a third time (e.g., **435**) that the charging station is reserved for the electric vehicle **105**. The third time can be earlier than or equal to the second time (e.g., **435**) at which the cooling system **420** is instructed to

activate.

(64) The one or more processors **310** can access a model (e.g., **425**) trained with machine learning and data (e.g., **430**) comprising values of the temperature of the component **410**, values of an ambient temperature (e.g., **430**) external to the charging station **405**, and metrics (e.g., **430**) indicating performance of charging sessions **475** of the charging station **405**. The metrics indicating performance can include any collection of data corresponding to the sessions **475** at the charging station **405**, such as rate of charge, power, current and voltage outputs, temperature readings and measurements, time to charge metrics, cooling rates and temperature measurements prior and post cooling, as well as any input **430** or output **435** data. The one or more processors **310** can determine, via the model (e.g., **425**) and based on the first time (e.g., **430**), the second time (e.g., **435**) at which to activate the cooling system **420**.

(65) The one or more processors **310** can identify an amount of power (e.g., **430**) to charge the electric vehicle **105**, identify a first value (e.g., **430**) of the temperature of the component measured by a sensor of the charging station and identify an ambient temperature (e.g., **430**) external to the charging station **405**. The one or more processors **310** can determine, via the model (e.g., **425**) and based on the amount of power (e.g., **430**), a second value (e.g., **435**) of the temperature of the component **410** configured to improve performance of charging relative to the first value (e.g., **430**). The one or more processors **310** can determine the second time (e.g., **435**) at which to activate the cooling system **420** and a rate of cooling of the cooling system **420** via the model (e.g., **425**) and based on the first time (e.g., **430**), the amount of power (e.g., **43**), the first value (e.g., **430**) of the temperature of the component, the second value (e.g., **435**) of the temperature of the component **410**, and the ambient temperature (e.g., **430**).

(66) In some aspects, system **400** relates to an electric vehicle charging station (e.g., **405**) that can include one or more processors **310** coupled with memory (e.g., **315**, **320**, **325**). The one or more processors **310** can receive, via a network **101**, an indication **415** from an electric vehicle **105** to establish a session **475** to charge the electric vehicle. The indication **415** can include a state of charge (e.g., **430**) of a battery (e.g., **110**, **115**, **120**) of the electric vehicle and a location (e.g., **430**) of the electric vehicle **105**. The one or more processors **310** can select, based on the state of charge of the battery and the location of the electric vehicle, a charging station **405** from a plurality of charging stations **405**. The one or more processors **310** can determine, responsive to the indication **415** and based on a distance between the electric vehicle and the charging station, a first time (e.g., **435**) at which the electric vehicle **105** arrives at the charging station **405** to charge the electric vehicle **105** in accordance with the session **475**. The one or more processors **310** can determine, based at least in part on the first time (e.g., **435**) and a temperature (e.g., **430**) of a component **410** of the charging station **405**, a second time (e.g., **435**) prior to arrival of the electric vehicle **105** at the charging station **405** at which to activate a cooling system **420** of the charging station **405**. The one or more processors **310** can provide an instruction **440** to activate the cooling system **420** at the second time to lower the temperature of the component **410** of the charging station **405** by the first time at which the electric vehicle **105** arrives at the charging station **405** to charge the electric vehicle **105**.

(67) In some aspects, the system **400** relates to a solution in which a system includes one or more processors coupled with memory. The one or more processors **310** can access a model (e.g., **425**) trained with machine learning and data (e.g., **430**) comprising one or more of: values of temperature of one or more components of a charging station, values of an ambient temperature external to the charging station, metrics indicating performance of charging sessions of the charging station and a distance of an electric vehicle from the charging station. The one or more processors **310** can identify, based on the model (e.g., **425**), a first time (e.g., **435**) at which an electric vehicle **105** arrives at a charging station **405** to charge the electric vehicle **105**. The one or more processors **310** can determine, based on at least one of the first time (e.g., **430**) and the temperature (e.g., **430**) of the component **410** input into the model (e.g., **425**), a second time (e.g.,

435) prior to arrival of the electric vehicle **105** at the charging station **405** at which to activate a cooling system **420** of the charging station **405**. The one or more processors **310** can provide an instruction **440** to activate the cooling system **420** at the second time to lower the temperature of the component **410** of the charging station **405** by the first time at which the electric vehicle **105** arrives at the charging station **405** to charge the electric vehicle **105**.

(68) FIG. 5 illustrates a method **500** to precool one or more charging stations **405**. The method **500** can be performed by one or more systems or components depicted in FIG. 1, FIG. 3 or FIG. 4, including, for example, a charging station, a data processing system or one or more processors. In brief overview, the method **500** can include receiving an indication at ACT **505**. At ACT **510**, the method **500** can include selecting a charging station. At ACT **515**, the method **500** can include determining EV arrival time. At ACT **520**, the method **500** can include determining the cooling time. At ACT **525**, the method **500** can include providing an instruction.

(69) Still referring to FIG. 5, among others, at ACT **505**, the method **500** can include receiving an indication. The method can receive, by the one or more processors via a network, an indication of a location of the electric vehicle. The one or more processors can receive, via a network, an indication from the electric vehicle to establish a session to charge the electric vehicle. The one or more processors can receive, via a sensor, an indication of a first value of the temperature of the component. The component can include a power cable or a cord, a power plug, power electronics unit or control box, or any electrical component of the charge station. The one or more processors can receive, via a sensor, an indication of an ambient temperature external to the charging station. The indication can include information about one or more temperature readings of one or more components of the charging station. The indication can include one or more temperature readings of the EV. The indication can include information about the location of the EV, speed of travel of the EV, traffic data corresponding to the area through which the EV is travelling to the charge station, state of charge of the battery of the EV, state of health of the EV battery, information about the EV, information about the EV driver, information about the account of the EV driver for paying the services for the EV charging. The indication can include information on the time at which EV is to arrive to the charge station, or the time that it is expected to take for the EV to arrive to the charge station.

(70) At ACT **510**, the method selects a charging station. The method can select, by the one or more processors, based on the location, the charging station from a plurality of charging station. The method can select the charging station based on the availability, quality, performance data or compatibility between the charging station and the EV. The charging station can be selected based on the temperature readings of one or more components of the EV. For example, the charging station can be selected because the one or more components of the charging station are cooler than other available charging stations. The charging station can be selected based on a determination that the amount of energy to cool the components of the charging station is lowest out of all available charging stations.

(71) At ACT **515**, the method identifies EV arrival time. The method can identify, by the one or more processors, a first time at which an electric vehicle arrives at a charging station to charge the electric vehicle. The method can identify the time of the EV arrival based on the indication identifying location of the EV and the location of the charging station. The method can identify the time of the EV arrival based on a determination of the distance between the EV and the charging station. The method can identify the time of the EV arrival based on a navigation function determining an estimated time of arrival of the EV to the charging station based on the current traffic reports and the shortest route for the EV to take to the charging station.

(72) At ACT **520**, the method determines the cooling time. The method can determine, by the one or more processors, based at least in part on the first time and a temperature of a component of the charging station, a second time prior to arrival of the electric vehicle at the charging station at which to activate a cooling system of the charging station. The method can determine, responsive

to the indication and based on a distance between the electric vehicle and the charging station, the first time at which the electric vehicle arrives at the charging station to begin the session. The one or more processors can compare a first value of temperature of a component of a charge station with a threshold established for the component. The one or more processors can determine, based at least in part on a comparison of the first value with the threshold, to activate the cooling system at the second time.

(73) The one or more processors can determine, based at least in part on the first time of EV arrival, the temperature of the component, and the ambient temperature external to the charging station, to activate the cooling system at the second time. The one or more processors can access a model trained with machine learning and data comprising values of the temperature of the component, values of an ambient temperature external to the charging station, and metrics indicating performance of charging sessions of the charging station. The one or more processors can determine, via the model and based on the first time, the second time at which to activate the cooling system.

(74) At ACT 525, the method provides instruction. The method can provide, by the one or more processors, an instruction to activate the cooling system at the second time to lower the temperature of the component of the charging station by the first time at which the electric vehicle arrives at the charging station to charge the electric vehicle. The one or more processors can establish, based on a distance between the selected charging station and the electric vehicle, a session to begin to charge the electric vehicle at the first time. The method can provide the instruction to cool a component of a charge station to a particular temperature level. The method can provide instruction to cool a first component of a charge station to a first temperature level or range and a second component of the charge station to a second temperature level or range. The method can provide the instruction in response to any one or more of: a make and model of the EV, state of the battery of the EV, a state of charge of the EV, an ambient temperature of the charge station, temperature of the EV battery, metrics of sessions of the charging station, temperature level of a component of the charging station and temperature of the water or fluid in the cooler or cooling system of the charging station.

(75) FIG. 6 illustrates an example method 600. The method 600 can include an ACT 605 of providing an electrical circuit of a charging station. The charging station can include or be electronically coupled to, directly or via a network, one or more processors coupled with memory. The charging station can identify, by the one or more processors coupled with memory, a first time at which an electric vehicle arrives at a charging station to charge the electric vehicle. The charging station can determine, by the one or more processors, based at least in part on the first time and a temperature of a component of the charging station, a second time prior to arrival of the electric vehicle at the charging station at which to activate a cooling system of the charging station. The charging station can provide, by the one or more processors, an instruction to activate the cooling system at the second time to lower the temperature of the component of the charging station by the first time at which the electric vehicle arrives at the charging station to charge the electric vehicle.

(76) Some of the description herein emphasizes the structural independence of the aspects of the system components or groupings of operations and responsibilities of these system components. Other groupings that execute similar overall operations are within the scope of the present application. Modules can be implemented in hardware or as computer instructions on a non-transient computer readable storage medium, and modules can be distributed across various hardware or computer based components.

(77) The systems described above can provide multiple ones of any or each of those components and these components can be provided on either a standalone system or on multiple instantiation in a distributed system. In addition, the systems and methods described above can be provided as one or more computer-readable programs or executable instructions embodied on or in one or more articles of manufacture. The article of manufacture can be cloud storage, a hard disk, a CD-ROM, a flash memory card, a PROM, a RAM, a ROM, or a magnetic tape. In general, the computer-

readable programs can be implemented in any programming language, such as LISP, PERL, C, C++, C #, PROLOG, or in any byte code language such as JAVA. The software programs or executable instructions can be stored on or in one or more articles of manufacture as object code. (78) Example and non-limiting module implementation elements include sensors providing any value determined herein, sensors providing any value that is a precursor to a value determined herein, datalink or network hardware including communication chips, oscillating crystals, communication links, cables, twisted pair wiring, coaxial wiring, shielded wiring, transmitters, receivers, or transceivers, logic circuits, hard-wired logic circuits, reconfigurable logic circuits in a particular non-transient state configured according to the module specification, any actuator including at least an electrical, hydraulic, or pneumatic actuator, a solenoid, an op-amp, analog control elements (springs, filters, integrators, adders, dividers, gain elements), or digital control elements.

(79) The subject matter and the operations described in this specification can be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. The subject matter described in this specification can be implemented as one or more computer programs, e.g., one or more circuits of computer program instructions, encoded on one or more computer storage media for execution by, or to control the operation of, data processing apparatuses. Alternatively or in addition, the program instructions can be encoded on an artificially generated propagated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal that is generated to encode information for transmission to suitable receiver apparatus for execution by a data processing apparatus. A computer storage medium can be, or be included in, a computer-readable storage device, a computer-readable storage substrate, a random or serial access memory array or device, or a combination of one or more of them. While a computer storage medium is not a propagated signal, a computer storage medium can be a source or destination of computer program instructions encoded in an artificially generated propagated signal. The computer storage medium can also be, or be included in, one or more separate components or media (e.g., multiple CDs, disks, or other storage devices include cloud storage). The operations described in this specification can be implemented as operations performed by a data processing apparatus on data stored on one or more computer-readable storage devices or received from other sources.

(80) The terms “computing device”, “component” or “data processing apparatus” or the like encompass various apparatuses, devices, and machines for processing data, including by way of example a programmable processor, a computer, a system on a chip, or multiple ones, or combinations of the foregoing. The apparatus can include special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). The apparatus can also include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, a cross-platform runtime environment, a virtual machine, or a combination of one or more of them. The apparatus and execution environment can realize various different computing model infrastructures, such as web services, distributed computing and grid computing infrastructures.

(81) A computer program (also known as a program, software, software application, app, script, or code) can be written in any form of programming language, including compiled or interpreted languages, declarative or procedural languages, and can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, object, or other unit suitable for use in a computing environment. A computer program can correspond to a file in a file system. A computer program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on

multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

(82) The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform actions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatuses can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Devices suitable for storing computer program instructions and data can include non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(83) The subject matter described herein can be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical user interface or a web browser through which a user can interact with an implementation of the subject matter described in this specification, or a combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network (“LAN”) and a wide area network (“WAN”), an inter-network (e.g., the Internet), and peer-to-peer networks (e.g., ad hoc peer-to-peer networks).

(84) While operations are depicted in the drawings in a particular order, such operations are not required to be performed in the particular order shown or in sequential order, and all illustrated operations are not required to be performed. Actions described herein can be performed in a different order.

(85) Having now described some illustrative implementations, it is apparent that the foregoing is illustrative and not limiting, having been presented by way of example. In particular, although many of the examples presented herein involve specific combinations of method acts or system elements, those acts and those elements may be combined in other ways to accomplish the same objectives. Acts, elements and features discussed in connection with one implementation are not intended to be excluded from a similar role in other implementations or implementations.

(86) The phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including” “comprising” “having” “containing” “involving” “characterized by” “characterized in that” and variations thereof herein, is meant to encompass the items listed thereafter, equivalents thereof, and additional items, as well as alternate implementations consisting of the items listed thereafter exclusively. In one implementation, the systems and methods described herein consist of one, each combination of more than one, or all of the described elements, acts, or components.

(87) Any references to implementations or elements or acts of the systems and methods herein referred to in the singular may also embrace implementations including a plurality of these elements, and any references in plural to any implementation or element or act herein may also embrace implementations including only a single element. References in the singular or plural form are not intended to limit the presently disclosed systems or methods, their components, acts, or elements to single or plural configurations. References to any act or element being based on any information, act or element may include implementations where the act or element is based at least in part on any information, act, or element.

(88) Any implementation disclosed herein may be combined with any other implementation or embodiment, and references to “an implementation,” “some implementations,” “one implementation” or the like are not necessarily mutually exclusive and are intended to indicate that a particular feature, structure, or characteristic described in connection with the implementation

may be included in at least one implementation or embodiment. Such terms as used herein are not necessarily all referring to the same implementation. Any implementation may be combined with any other implementation, inclusively or exclusively, in any manner consistent with the aspects and implementations disclosed herein.

(89) References to “or” may be construed as inclusive so that any terms described using “or” may indicate any of a single, more than one, and all of the described terms. References to at least one of a conjunctive list of terms may be construed as an inclusive OR to indicate any of a single, more than one, and all of the described terms. For example, a reference to “at least one of ‘A’ and ‘B’” can include only ‘A’, only ‘B’, as well as both ‘A’ and ‘B’. Such references used in conjunction with “comprising” or other open terminology can include additional items.

(90) Where technical features in the drawings, detailed description or any claim are followed by reference signs, the reference signs have been included to increase the intelligibility of the drawings, detailed description, and claims. Accordingly, neither the reference signs nor their absence have any limiting effect on the scope of any claim elements.

(91) Modifications of described elements and acts such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations can occur without materially departing from the teachings and advantages of the subject matter disclosed herein. For example, elements shown as integrally formed can be constructed of multiple parts or elements, the position of elements can be reversed or otherwise varied, and the nature or number of discrete elements or positions can be altered or varied. Other substitutions, modifications, changes and omissions can also be made in the design, operating conditions and arrangement of the disclosed elements and operations without departing from the scope of the present disclosure.

(92) For example, a computer system **300** described in FIG. **3** can be used in conjunction with, instead of, or together with systems **400** or **500** or their system components, and vice versa. Further relative parallel, perpendicular, vertical or other positioning or orientation descriptions include variations within $\pm 10\%$ or ± 10 degrees of pure vertical, parallel or perpendicular positioning. References to “approximately,” “substantially” or other terms of degree include variations of $\pm 10\%$ from the given measurement, unit, or range unless explicitly indicated otherwise. Coupled elements can be electrically, mechanically, or physically coupled with one another directly or with intervening elements. Scope of the systems and methods described herein is thus indicated by the appended claims, rather than the foregoing description, and changes that come within the meaning and range of equivalency of the claims are embraced therein.

Claims

1. A system, comprising: one or more processors, coupled with memory, to: identify a first time at which an electric vehicle arrives at a charging station to charge the electric vehicle; determine, based at least in part on the first time and a temperature of a component of the charging station, a second time prior to arrival of the electric vehicle at the charging station at which to activate a cooling system of the charging station; and provide an instruction to activate the cooling system at the second time to lower the temperature of the component of the charging station by the first time at which the electric vehicle arrives at the charging station to charge the electric vehicle.
2. The system of claim 1, comprising the one or more processors to: receive, via a network, an indication from the electric vehicle to establish a session to charge the electric vehicle; and determine, responsive to the indication and based on a distance between the electric vehicle and the charging station, the first time at which the electric vehicle arrives at the charging station to begin the session.
3. The system of claim 1, comprising the one or more processors to: receive, via a network, an indication of a location of the electric vehicle; select, based on the location, the charging station

from a plurality of charging station; and establish, based on a distance between the selected charging station and the electric vehicle, a session to begin to charge the electric vehicle at the first time.

4. The system of claim 1, comprising the one or more processors to: receive, via a sensor, an indication of a first value of the temperature of the component, the component comprising at least one of a power cable or power electronics; compare the first value with a threshold established for the component; and determine, based at least in part on a comparison of the first value with the threshold, to activate the cooling system at the second time.

5. The system of claim 1, comprising the one or more processors to: receive, via a sensor, an indication of an ambient temperature external to the charging station; and determine, based at least in part on the first time, the temperature of the component, and the ambient temperature external to the charging station, to activate the cooling system at the second time.

6. The system of claim 1, comprising the one or more processors to: receive, from the electric vehicle via a network, an indication of a state of charge of a battery of the electric vehicle and a location of the electric vehicle; select, based on the state of charge of the battery and the location of the electric vehicle, the charging station from a plurality of charging stations; and predict, based on a distance between the selected charging station and the electric vehicle, the first time at which the electric vehicle arrives at the charging station.

7. The system of claim 1, comprising the one or more processors to: receive, from the electric vehicle via a network, an indication of a state of charge of a battery of the electric vehicle; determine, based on the state of charge, an amount of power to charge the battery of the electric vehicle; and establish, based on the amount of power to charge the battery, to activate the cooling system at the second time.

8. The system of claim 7, comprising the one or more processors to: receive a first value of the temperature of the component measured via a sensor; determine, based on the amount of power to charge the battery, a second value of the temperature for the component less than the first value that increases a rate of charge of the charging station relative to the first value of the temperature; and determine the second time at which to activate the cooling system based on the first value of the temperature, the second value of the temperature, and the first time of arrival.

9. The system of claim 8, comprising the one or more processors to: establish a rate of cooling of the cooling system based on the first value of the temperature, the second value of the temperature, and the first time of arrival; and provide the instruction to activate the cooling system to cool at the established rate of cooling.

10. The system of claim 1, comprising the one or more processors to: instruct the charging station to present a notification at a third time that the charging station is reserved for the electric vehicle, the third time earlier than or equal to the second time at which the cooling system is instructed to activate.

11. The system of claim 1, comprising the one or more processors to: identify, based at least on an indication of location of the electric vehicle input into a model trained via machine learning, the first time; and determine, based at least in part on the first time and the temperature of the component of the charging station input into the model, the second time.

12. The system of claim 1, the one or more processors to: access a model trained with machine learning and data comprising values of the temperature of the component, values of an ambient temperature external to the charging station, and metrics indicating performance of charging sessions of the charging station; and determine, via the model and based on the first time, the second time at which to activate the cooling system.

13. The system of claim 12, comprising the one or more processors to: identify an amount of power to charge the electric vehicle; identify a first value of the temperature of the component measured by a sensor of the charging station; identify an ambient temperature external to the charging station; determine, via the model and based on the amount of power, a second value of the temperature of

the component configured to improve performance of charging relative to the first value; and determine the second time at which to activate the cooling system and a rate of cooling of the cooling system via the model and based on the first time, the amount of power, the first value of the temperature of the component, the second value of the temperature of the component, and the ambient temperature.

14. A method, comprising: identifying, by one or more processors coupled with memory, a first time at which an electric vehicle arrives at a charging station to charge the electric vehicle; determining, by the one or more processors, based at least in part on the first time and a temperature of a component of the charging station, a second time prior to arrival of the electric vehicle at the charging station at which to activate a cooling system of the charging station; and providing, by the one or more processors, an instruction to activate the cooling system at the second time to lower the temperature of the component of the charging station by the first time at which the electric vehicle arrives at the charging station to charge the electric vehicle.

15. The method of claim 14, comprising: receiving, by the one or more processors via a network, an indication from the electric vehicle to establish a session to charge the electric vehicle; and determining, by the one or more processors, responsive to the indication and based on a distance between the electric vehicle and the charging station, the first time at which the electric vehicle arrives at the charging station to begin the session.

16. The method of claim 14, comprising: receiving, by the one or more processors via a network, an indication of a location of the electric vehicle; selecting, by the one or more processors, based on the location, the charging station from a plurality of charging station; and establishing, by the one or more processors, based on a distance between the selected charging station and the electric vehicle, a session to begin to charge the electric vehicle at the first time.

17. The system of claim 14, comprising: receiving, by the one or more processors via a sensor, an indication of a first value of the temperature of the component, the component comprising at least one of a power cable or power electronics; comparing, by the one or more processors, the first value with a threshold established for the component; and determining, by the one or more processors, based at least in part on a comparison of the first value with the threshold, to activate the cooling system at the second time.

18. The method of claim 14, comprising: receiving, by the one or more processors via a sensor, an indication of an ambient temperature external to the charging station; and determining, by the one or more processors, based at least in part on the first time, the temperature of the component, and the ambient temperature external to the charging station, to activate the cooling system at the second time.

19. The method of claim 14, comprising: accessing, by the one or more processors, a model trained with machine learning and data comprising values of the temperature of the component, values of an ambient temperature external to the charging station, and metrics indicating performance of charging sessions of the charging station; and determining, by the one or more processors, via the model and based on the first time, the second time at which to activate the cooling system.

20. An electric vehicle charger comprising one or more processors, coupled with memory, to: receive, via a network, an indication from an electric vehicle to establish a session to charge the electric vehicle, the indication comprising a state of charge of a battery of the electric vehicle and a location of the electric vehicle; select, based on the state of charge of the battery and the location of the electric vehicle, a charging station from a plurality of charging stations; determine, responsive to the indication and based on a distance between the electric vehicle and the charging station, a first time at which the electric vehicle arrives at the charging station to charge the electric vehicle in accordance with the session; determine, based at least in part on the first time and a temperature of a component of the charging station, a second time prior to arrival of the electric vehicle at the charging station at which to activate a cooling system of the charging station; and provide an instruction to activate the cooling system at the second time to lower the temperature of the

component of the charging station by the first time at which the electric vehicle arrives at the charging station to charge the electric vehicle.
